DNS

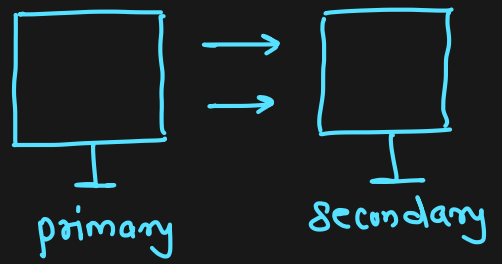
DNS

→ resolver
→ root DNS

→ high availability → replication

- The Domain Name System (DNS) is a hierarchical and distributed naming system for computers, services, and other resources in the Internet or other Internet Protocol (IP) networks
- It associates various information with domain names assigned to each of the associated entities
- Most prominently, it translates readily memorized domain names to the numerical IP addresses needed for locating and identifying computer services and devices with the underlying network protocols
- The Domain Name System has been an essential component of the functionality of the Internet since 1985 →
- Uses port no 53
- **Why DNS is Needed**
  - Humans remember names, not IP addresses
  - IP addresses can change; names remain stable
  - Enables scalability, decentralization, and load distribution

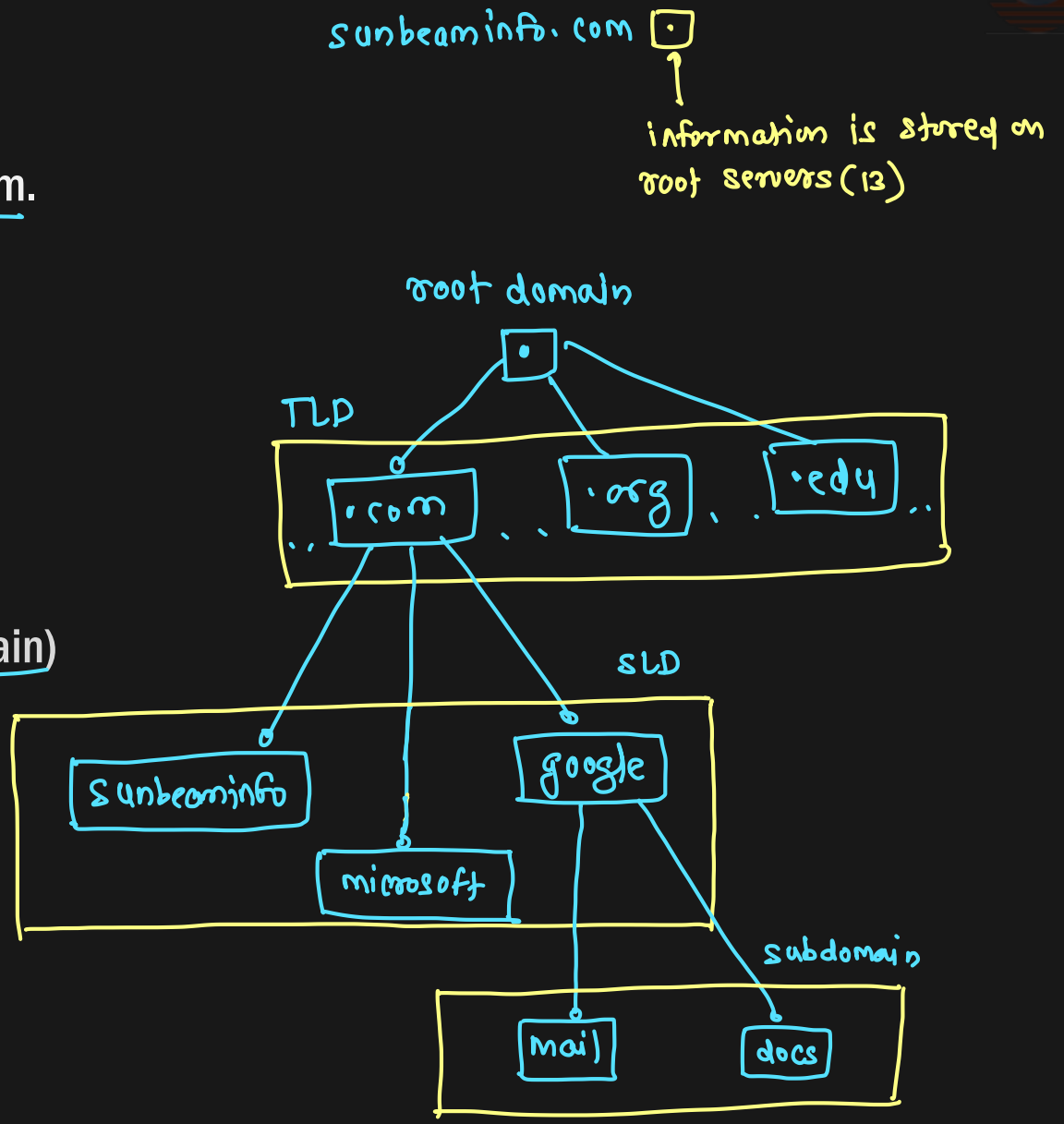
translate → Domain name → IP address (public network)
translate → Hostname → IP address (private network)



linux/macos : /etc/hosts
windows : c:\windows\system32\drivers\etc\hosts

Hierarchy of Domain Names

- **Root Domain:**
  - The dot (.) at the end (often invisible). Example: example.com.
- **Top-Level Domain (TLD):**
  - .com, .org, .net, .uk, .edu
- **Second-Level Domain (SLD)**
  - The registrable part. Example: google in google.com
- **Subdomain**
  - Optional prefix. Example: mail.google.com (mail is subdomain)



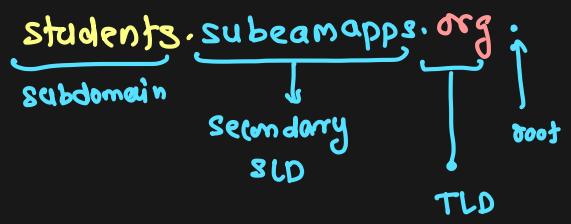
Terminologies



- **DNS Root Servers** → stores information about TLD servers
  - Name server for root zone of DNS of the internet
  - It directly answers requests for records in the root zone and answers other requests by returning a list of the authoritative name servers for the appropriate Top Level Domain (TLD)
  - TLDs are controlled by the root zone

- **Top Level Domain (TLD)**
  - A top-level domain (TLD) is one of the domains at the highest level in the hierarchical Domain Name System of the Internet
  - IANA currently distinguishes the following groups of top-level domains:
    - country-code top-level domains (ccTLD) → .co.in, .in, .uk, .us
    - generic top-level domains (gTLD)
      - sponsored top-level domains (sTLD) → .com
      - unsponsored top-level domains → .edu
    - infrastructure top-level domain (.arpa)

Terminologies



- **Subdomain**
  - Anything between the host and TLS is known as subdomain
  - E.g. sunbeaminfo.com is subdomain of .com TLD

- **Zone and zone files** → database of domain name to IP address mapping
  - A zone of zone file is a mapping of IP address and the hosts for a given subdomain
  - The zone file for sunbeaminfo.com would contain a record for www

- **Name Server** → machine with DNS server configured
  - A server that runs the DNS service and can either store or cache the information for the DNS platform
  - Whether it stores or caches depends on whether it is referred from a higher level

- **Authoritative**
  - The organizations run name servers that become authoritative
  - They can answer the queries with authority
    - eg. GoDaddy.com

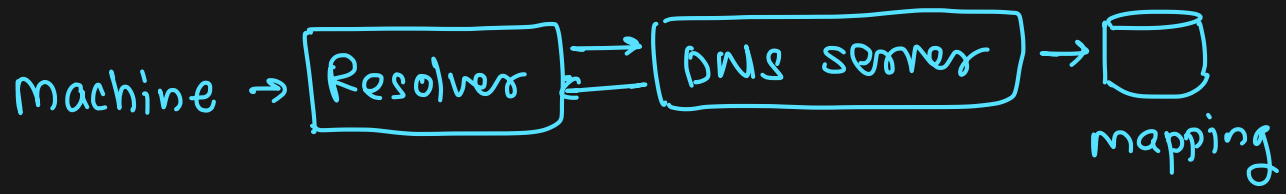
Terminologies



- **Resource Record (RR)** → stored within the zones
  - This is information of element for network resources
  - The types of information elements are categorized and organized with a list of DNS record types
  - Resource records of the same type are described as a *resource record set* (RRset)
  - DNS resolvers return the entire set upon query, but servers may implement round-robin ordering to achieve load balancing
  - Each record has a
    - NAME: Name of the node to which this record pertains
    - TYPE: Type of RR in numeric form (e.g., 15 for MX RRs)
    - CLASS: Class code
    - TTL: Count of seconds that the RR stays valid (The maximum is $2^{31}-1$, which is about 68 years)
    - RDATA: Additional RR-specific data
    - RDLENGTH: Length of RDATA field (specified in octets)

Terminologies

- **Hosts**
  - A record in the zone file: www



- **FQDN**
  - Fully Qualified Domain Name
  - The host and the domain name

- **Registrar** : godaddy | AWS | GCP
  - A domain name registrar is a company that manages the reservation of Internet domain names
  - A domain name registrar must be accredited by a generic top-level domain (gTLD) registry or a country code top-level domain (ccTLD) registry
  - A registrar operates in accordance with the guidelines of the designated domain name registries

- **Resolver**
  - The client side of the DNS is called a DNS resolver
  - A resolver is responsible for initiating and sequencing the queries that ultimately lead to a full resolution (translation) of the resource sought, e.g., translation of a domain name into an IP address

Types of DNS Records → resource records

redirection

domain1.com → domain2.com

- **A (Address):** Maps domain to IPv4 address
- **AAAA:** Maps domain to IPv6 address
- **CNAME (Canonical Name):** Aliases one domain to another (e.g., www to @)
- **MX (Mail Exchange):** Directs email to mail servers (with priority values)
- **TXT (Text):** Holds human/machine-readable text (used for verification)
- **NS (Name Server):** Specifies the authoritative DNS servers for the domain
- **PTR (Pointer):** Maps IP to domain (reverse DNS lookup)
- **SOA (Start of Authority):** Contains administrative info about the zone

DNS Zone : database of RRs

- It is a **portion of the DNS namespace** that is managed, administered, and stored by a single organization or administrator
- It is the **administrative boundary** for DNS data
- A zone is NOT the same as a domain
  - A domain is a node in the namespace; a zone is the part of the namespace that a particular server is authoritative for
- A zone contains **resource records** stored in a **zone file** on an authoritative name server
- **Why Zones Exist**
  - DNS is distributed — no single server holds all records
  - Zones **delegate authority**: an organization controls its own zone without asking the root or TLD servers every time
  - Zones enable **local management**, **redundancy**, and **scalability**
- **Zone vs Domain**

A zone can store multiple domain's information

Domain	Zone
A node in the DNS namespace	A portion of namespace administered together
Logical concept	Administrative concept
example.com is a domain	example.com may be one or more zones
Can contain many subdomains	Contains records for names within its boundary

Types of DNS Zones

- **Forward Zone (Forward Lookup Zone)**
  - Maps **domain names → IP addresses**
  - Most common type
  - Contains A, AAAA, CNAME, MX, TXT, etc
  - E.g. www.example.com → 93.184.216.34
- **Reverse Zone (Reverse Lookup Zone)**
  - Maps **IP addresses → domain names**
  - Uses **PTR records**
  - Uses special domain in-addr.arpa (IPv4) and ip6.arpa (IPv6)
  - E.g. 34.216.184.93.in-addr.arpa → www.example.com
- **Primary Zone (Master Zone)**
  - The **writable, authoritative copy** of a zone
  - Edits are made here first
  - Hosted on the **primary name server**
- **Secondary Zone (Slave Zone)**
  - **Read-only copy** of the primary zone
  - Gets updates via **zone transfer** (AXFR/IXFR)
  - Provides **redundancy** and **load balancing**
  - Cannot be edited directly

Zone File Components



- Every zone file must have
  - SOA (Start of Authority) — one per zone; defines zone parameters
  - NS records — authoritative servers for the zone
  - A/AAAA records — host addresses
  - Other records — MX, CNAME, TXT, etc
- SOA Record Fields
  - Primary NS: Master server for the zone
  - Responsible person: Admin email (with . instead of @)
  - Serial: Version number; incremented on changes
  - Refresh: How often secondary checks for updates
  - Retry: Wait time before retrying failed refresh
  - Expire: When secondary stops answering if no refresh
  - Minimum TTL: Default TTL for records / negative caching

Types of DNS Servers



- **Authoritative DNS Servers**
  - In **authoritative server** is the **final source of truth** for a specific zone
  - It holds the actual DNS records (A, AAAA, MX, etc.) and gives definitive answers
  - **Characteristics**
    - Returns answers with the **AA (Authoritative Answer)** flag set
    - Does **not** query other servers on behalf of clients
    - Only answers for zones it is configured to manage
    - Can be queried directly by resolvers
- **Recursive DNS Servers (Resolvers)**
  - A **recursive resolver** does the "legwork" of finding answers for clients
  - It queries multiple servers (root → TLD → authoritative) and returns the final answer
  - **Characteristics**
    - Accepts queries from clients (stub resolvers)
    - **Recursively** queries other servers to find the answer
    - **Caches** results for future queries (based on TTL)
    - Does **not** hold authoritative data (unless combined with authoritative role)
    - Returns answers with the **RA (Recursion Available)** flag

Type of DNS Server

- **Root Name Servers**
  - The **root servers** sit at the very top of the DNS hierarchy
  - **Characteristics**
    - **13 logical root server clusters** (named A through M)
    - Hundreds of physical instances worldwide via **anycast**
    - Managed by ICANN and various organizations (Verisign, NASA, etc.)
    - Know the addresses of all **TLD name servers**
    - Do **not** know individual domain IPs — only where to find TLD servers
  - **Role:** When a resolver has no cache, it asks a root server: "Where is .com?" — the root replies with the .com TLD server addresses
- **TLD (Top-Level Domain) Name Servers**
  - These servers handle **top-level domains** like .com, .org, .net, .in, .uk.
  - **Characteristics**
    - Maintained by **TLD registries** (e.g., Verisign for .com)
    - Know the **authoritative name servers** for every domain under their TLD
    - Return **referrals** to authoritative servers
  - **Types of TLDs**
    - **gTLD** — Generic (.com, .org, .net, .info)
    - **ccTLD** — Country code (.in, .uk, .jp)
    - **Sponsored TLD** — .edu, .gov, .mil
    - **New gTLD** — .app, .dev, .xyz

How DNS Resolution Works (8 Steps)

- **User types** example.com into browser
- **Check cache:** Browser to OS to Router to ISP Resolver (if found, skip to end)
- **Query Root Server:** Resolver asks root "Where is .com?" Root replies with TLD server address
- **Query TLD Server:** Resolver asks .com server "Where is example.com?" TLD replies with Authoritative server
- **Query Authoritative Server:** Resolver asks the domain's own DNS server "What is the IP of example.com?"
- **Response:** Authoritative server returns A record (IP address)
- **Caching:** Resolver caches IP for future queries (based on TTL)
- **Deliver:** Resolver sends IP to browser, browser loads website

